

Akanksha Mishra

Doctoral Researcher (Ph.D.) | Machine Learning & Artificial Intelligence
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RESEARCH PROFILE

My research focuses on Bayesian and probabilistic machine learning, deep latent-variable models, interpretable AI, and feature selection, with applications to high-dimensional biomedical data and decision-making. My doctoral research investigates Bayesian approaches for learning latent structure for feature selection and preference modeling. I am particularly interested in developing machine-learning methods that are interpretable, uncertainty-aware, and robust for complex high-dimensional data, with applications including cancer genomics.

RESEARCH INTERESTS

Bayesian & Probabilistic Machine Learning, Deep Latent Variable Models, Feature Selection, Deep Learning, Preference Learning & Decision Making, ML for Cancer Genomics.

EDUCATION

- **Doctor of Philosophy (Ph.D.)**[Thesis Submitted: July 2026] 2026
School of Mathematics and Computer Science, IIT Goa
Thesis Advisor: Prof. Clint Pazhayidam George
Thesis Title: Bayesian Learning of Latent Structure for Feature Selection and Preference Modeling
- **Master of Technology (M.Tech.)** 2015
CSE, IIT Delhi, India
- **Bachelor of Technology (B.Tech.)** 2012
CSE, Uttar Pradesh Technical University, India

EXPERIENCE

- **Research Fellow** December 2017–December 2019
CSE, IIT (BHU) Varanasi
- **Teaching Fellow / Co-instructor** January 2017–December 2017
CSE, IIT Delhi
- **Assistant Professor** July 2015–December 2016
CSE, BML Munjal University
- **Research Associate and Member** July 2014–July 2015
Software Analytics Lab, IIT Delhi

SELECTED RESEARCH CONTRIBUTIONS

Bayesian Feature Selection for Cancer Genomics

Developed a weight-sharing Bayesian neural network for consistent feature selection, with applications to high-dimensional cancer gene-expression data. This work combines Bayesian learning and deep neural networks to address feature-selection challenges in biomedical data.

Bayesian Learning for Preference Modeling

Developed an interactive multi-objective decision-making framework based on Bayesian learning of hidden preferences, addressing preference elicitation and decision-making under uncertainty.

Text Analysis for Online Harm and Social Media

Contributed to research on text analysis for detecting and preventing online harm, including cyberbullying, hate speech, and offensive content in online communications.

PUBLICATIONS/PREPRINTS

- [1] Akanksha Mishra, Wei Xia, and Clint Pazhayidam George. A weight-sharing bayesian neural network for consistent feature selection with applications in cancer gene expression data. *BMC bioinformatics*, 2026.
- [2] Akanksha Mishra, Wei Xia, and Clint Pazhayidam George. Interactive multi-objective decision making through bayesian learning of hidden preferences. (*Under review*), 2026.
- [3] Akanksha Mishra, Sharad Sinha, and Clint Pazhayidam George. Shielding against online harm: A survey on text analysis to prevent cyberbullying. *Engineering Applications of Artificial Intelligence*, 133:108241, 2024.

- [4] Harshit Mehrotra, Akanksha Mishra, and Sukomal Pal. A multi-stage classification framework for disaster-specific tweets. *SN Computer Science*, 3(1):24, 2022.
- [5] Akanksha Mishra and Sukomal Pal. IIT varanasi at HASOC 2019: Hate speech and offensive content identification in indo-european languages. 2019.
- [6] Akanksha Mishra and Sukomal Pal. IIT BHU at TREC 2019 incident streams track. 2019.
- [7] Akanksha Mishra and Sukomal Pal. IIT-BHU at CIQ 2019: Classification of insincere questions. 2019.
- [8] Akanksha Mishra, Rajesh Kumar Mundotiya, and Sukomal Pal. IIT (BHU)@ IECSIL-FIRE2018: language independent automatic framework for entity extraction in indian languages. 2018.
- [9] Akanksha Mishra and Richa Sharma. Enriching frame-based structured representations for requirements using case frames-an approach towards handling incompleteness in informal requirements. In *International Conference on Evaluation of Novel Approaches to Software Engineering*, volume 2, pages 314–319. SCITEPRESS, 2017.
- [10] Sunil Gupta, Akanksha Mishra, and Mukesh Chawla. Analysis and recommendation of common fault and failure in software development systems. In *2016 International conference on signal processing, communication, power and embedded system (SCOPES)*, pages 1730–1734. IEEE, 2016.
- [11] Akanksha Mishra and Ashish Sureka. A graph processing based approach for automatic detection of semantic inconsistency between bpmn process model and sbvr rules. In *International Conference on Mining Intelligence and Knowledge Exploration*, pages 115–129. Springer, 2015.

INVITED TALKS

- **Indian Symposium on Machine Learning (IndoML)**, 21st – 23rd Dec 2023, IIT Bombay, India. **Title of the poster presentation:** Feature Selection using Deep Learning based models
- **Workshop on “The role of AI in transforming healthcare”**, 11th – 12th June 2022, Jointly organized by the Translational Health Science Technology Institute (THSTI), University of Oxford, and the Centre for Integrative Biology and Systems Medicine (IBSE), IIT Madras, India. **Title of the talk and poster presentation:** Deep Learning and Probabilistic Approaches for Cancer Genomics

REVIEWER

Workshop: “Structured Data for Health” Workshop at ICML 2026, Women in Machine Learning workshop at NeurIPS 2026.

RELEVANT COURSES

Machine Learning, Optimization, Artificial Intelligence, Pattern Recognition, Information Retrieval, Natural Language Processing, and Deep Learning.

ACHIEVEMENTS

- Travel/Fellowship Grants: IndoML (2026, 2024, 2023), ICML (2022), ICLR (2022), AISTATS (2022), NeurIPS (2021), CODS-COMAD (2020), FIRE (2019).
- UGC-NET (2018); GATE (2013).

POSITION OF RESPONSIBILITY

- **Coordinator**, Research Scholars Confluence, IIT Goa, 2022
- **Vice Chair**, ACM Student Chapter, IIT Goa, 2021-2022
- **Teaching Assistant**, AI Shiksha — Introduction to Machine Learning, National Supercomputing Mission, 2021
- **Member**, ACM and IEEE

TEACHING ASSISTANT

- **Machine Learning**, IIT Goa, Spring 2021, 2022, 2025 and 2026.
- **Probability and Statistics**, IIT Goa, Autum 2024.
- **Software Tools**, IIT Goa, Spring 2024.
- **Introduction to Computing**, IIT Goa, Autumn 2023.
- **Artificial Intelligence**, IIT Goa, Autumn 2020, 2021 and 2022